

# Combinatorial analysis

## Chapter outline

1. Introduction
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7. Permutation (with repetition) - Multinomial formula.
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In this course,  $\mathbb{N}$  denotes the set of positives integers.

## 1.1 Introduction

**Combinatorial analysis** is the set of techniques used to count (enumerate) the dispositions that can be formed from the elements of a finite set (or the outcomes of a given experiment).

**Example 1.1.1.** 1. *In a first year class, there are 4 groups of 25 students. How many different choices are there to form a delegation of 4 students, (one student per group)?*

2. *How many 5-digit number plates can we form?*
3. *In statistics, 15 students are admitted to the postgraduate competition. How many ranking can we have if we assume that two students can not get the same ranking?*
4. *We put 5 people in a car, how many choices are there?*

When trying to count all the possible dispositions, two things need to be clear:

- The structure of the set of elements or the results of experiment (discernible or indiscernible elements).

- The structure of dispositions (ordered or unordered dispositions, with or without repetition)

**Definition 1.1.2.** Two elements of a set are said to be **indiscernible** (or indistinguishable) if they are considered equivalent, otherwise they are said to be **discernible** (or distinguishable).

**Example 1.1.3.** The elements of 32-card deck are:

- discernible two by two, if we consider the color: clubs, diamonds, hearts, spades; and the height: 7, 8, 9, 10, jack, queen, king, ace.
- composed of 8 groups of 4 elements indiscernible, if we consider that two cards of the same height are equivalent.
- composed of 4 groups of 8 elements indiscernible, if we consider that two cards of the same color are equivalent.

**Definition 1.1.4.** We call a **disposition**, the elements selected from a set. If an element appears more than once, it is called a disposition **with repetition**, otherwise it is called a disposition **without repetition**. If the elements in the same disposition play the same role, it is said that the disposition is **unordered** otherwise it is **ordered**.

**Example 1.1.5.**

In an urn, there are 26 balls of the same size and color; on each ball an alphabet letter is written; 3 balls are drawn:

1. one by one and put the drawn ball back into the urn, in this order we get the word: ZOO. ZOO is an ordered disposition with repetition;
2. one by one and the drawn ball is not put back into the urn, we get the word: LIT. LIT an ordered disposition without repetition;
3. at the same time, we get the letter group  $\{A, B, C\}$ .  $\{A, B, C\}$  is a unordered disposition without repetition;
4. one by one by putting back the ball in the urn and we are interested in the group of letters obtained, namely:  $\{R, R, S\}$ .  $\{R, R, S\}$  is a unordered disposition with repetition.

In this chapter, the classic dispositions will be studied: arrangements, permutations, combinations; (with and without repetition).

In what follows,  $E$  denotes a set of  $n$  elements (ie:  $\text{card } E = |E| = n$ ) or a set of results of an experiment with  $n$  outcomes.

## 1.2 Fundamental principal of counting - Arrangement with repetition

**Theorem 1.2.1** (Fundamental principal of counting).

*Two consecutive experiments are realized, the first one has  $n$  possibilities, and the second one has  $m$  possibilities for each possibility of the first experiment. Then there are  $n \cdot m$  possibilities for the two experiments taken together.*

*Proof.* We denote the  $n$  possibilities of the first experiment by  $a_1, a_2, \dots, a_n$  and the  $m$  possibilities of the second experiment by  $b_1, b_2, \dots, b_m$ .

if we denote  $(a, b)$  one of the possibilities of the two experiments taken together,  $a$  is a possibility of the first experiment and  $b$  is for the second one, we get:

$$\underbrace{\left. \begin{array}{c} (a_1, b_1); (a_1, b_2); \dots; (a_1, b_m) \\ (a_2, b_1); (a_2, b_2); \dots; (a_2, b_m) \\ \vdots \\ (a_n, b_1); (a_n, b_2); \dots; (a_n, b_m) \end{array} \right\}^n}_m$$

Then, there are  $n \cdot m$  possibilities for the two experiments taken together. □

**Remark 1.2.2.** *The result of the two experiments taken together is called **couple** or a **sequence of length 2**.*

**Corollary 1.2.3** (Generalized fundamental principal).

*$k$  experiments are realized. The first one has  $n_1$  possibilities, the seconde has  $n_2$  possibilities for each possibility of the first experiment, the third has  $n_3$  possibilities for each possibility of the two experiments taken together,  $\dots$ , the  $k^{\text{st}}$  has  $n_k$  possibilities for each possibility of the  $k - 1$  experiments taken together. Then, there are  $n_1 \cdot n_2 \cdot \dots \cdot n_k = \prod_{i=1}^k n_i$  possibilities for  $k$  experiments taken together.*

*Proof.* The proof is done by induction on  $k$ . □

**Remark 1.2.4.** *The result of  $k$  experiments taken together is called  **$k$ -tuple** or a **sequence of length  $k$** .*

**Corollary 1.2.5.**  *$k$  identical experiments are realized, and each one of them has  $n$  possibilities. Then there are  $n^k$  possibilities for  $k$  experiments taken together.*

*Proof.* In corollary 2.3, we put:  $n_1 = n_2 = \dots = n_k = n$ . □

This corollary leads us to the following definition

**Definition 1.2.6.**

*We call an **arrangement with repetition** of  $n$  elements, a sequence of length  $k$  in which the terms are distinct elements or not, (or it is an ordered disposition with repetition, of  $k$  elements chosen from  $n$ ), this number is denoted  $P_n^k$ .*

In the same arrangement with repetition, some of the elements can appear several times.

**Corollary 1.2.7.** By corollary 2.5, we have:  $P_n^k = n^k$ .

**Example 1.2.8.**

1. For the 5<sup>th</sup> year engineering students, 9 subjects are proposed, for each subject there are 3 interested pairs. We say that a pair is satisfied if it gets the subject that they are interested in. How many pairs are satisfied?

**Answer:** We consider that the different subjects as the first experiment and the choice of a pair as second experiment, then by fundamental principal of counting (Theorem 1.2.1), there are  $9 \cdot 3 = 27$  possible choices.

2. In O.R., the student committee consists of: 8 students of 1<sup>st</sup> year, 6 students of 2<sup>nd</sup> year, 5 students of 3<sup>rd</sup> year, 4 students of 4<sup>th</sup> year, and 2 students of 5<sup>th</sup> year (in all 25 students). We want to choose a sub-committee of 5 students with one representative per year. How many sub-committees can be formed?

**Answer:** We assume that the choice of a sub-committee is the combined result of 5 distinct experiments, so by the generalized fundamental principal of counting (Corollary 1.2.3), there are  $8 \cdot 6 \cdot 5 \cdot 4 \cdot 2 = 1920$ , choices of sub-committees.

3. A coin is tossed  $n$  times, how many possible results can we get?

**Answer:** There are two possible outcomes for each roll: heads or tails, moreover this experiment is performed  $n$  times, so by Corollary 1.2.5, there are  $P_2^n = 2^n$  possible results. (It is the number of arrangements with repetition of 2 elements taken  $n$  to  $n$ ).

4. How many 7-digit plates can we form, if the two starting digits are letters and the 5 last ones are numbers?

**Answer:** By the generalized fundamental principal, the answer is:  $26^2 10^5 = 67600000$ .

## 1.3 Arrangement (without repetition)

**Definition 1.3.1.** We call an **arrangement** without repetition (we often do not specify **without repetition**) of  $n$ , a sequence of length  $k$  in which the terms are distinct elements, (or it is an ordered disposition without repetition, of  $k$  elements chosen from  $n$ ), this number is denoted  $(n)_k$ .

In a such arrangement (without repetition), each one of the  $k$  elements appear in a single fixed order, then  $1 \leq k \leq n$ .

**Theorem 1.3.2.**

$$(n)_k = n(n-1)(n-2) \cdots (n-k+1) \quad 1 \leq k \leq n \quad (1.3.1)$$

*Proof.* We consider  $k$  identical experiments that do not reproduce the same result. the 1<sup>st</sup> experiment has  $n$  possible results,  
the 2<sup>nd</sup> experiment has  $(n - 1)$  possible results,  
the 3<sup>rd</sup> experiment has  $(n - 2)$  possible results.

$\vdots$

the  $k^{\text{st}}$  experiment has  $(n - k + 1)$  possible results.

Using Corollary 1.2.3 with,  $n_1 = n, n_2 = n - 1, n_3 = n - 2, \dots, n_k = n - k + 1$ , so the number of possible results of the  $k$  experiments is  $(n)_k = n(n - 1)(n - 2) \cdots (n - k + 1)$ .  $\square$

**Remark 1.3.3.** 1. We can extend the definition of  $(n)_k$  for  $k \in \mathbb{N} \cup \{0\}$

$$(n)_k = \begin{cases} 1, & \text{if } k = 0, \\ n(n - 1) \cdots (n - k + 1), & \text{if } k = 1, 2, \dots, n, \\ 0, & \text{if } k > n. \end{cases} \quad (1.3.2)$$

2. We introduce the factorial notation: for an integer  $n \geq 0$ ,

$$n! = n(n - 1) \cdots 2 \cdot 1 = n \cdot (n - 1)!, \text{ we say "n factorial", with the convention } 0! = 1.$$

The relation (1.3.2) becomes:

$$(n)_k = \frac{n!}{(n - k)!} \quad k = 0, 1, 2, \dots, n. \quad (1.3.3)$$

Some values of  $n!$ :  $0! = 1, 1! = 1, 2! = 2, 3! = 6, 4! = 24, 5! = 120, 6! = 720, 7! = 5040, 8! = 40320, 9! = 362880$ .

**Example 1.3.4.**

1. How many 5-digit number plates can we form if no digit is repeated?

**Answer:** it is the number of arrangements of 5 elements taken from 10, so there are  $(10)_5 = \underbrace{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6}_{\text{there are 5 terms}} = 30240$  plates.

2. How many arrangements can be made with a letter and a parcel in 9 letterboxes numbered from 1 to 9, without putting them together?

**Answer:** it is the number of arrangements of 2 elements taken from 9, so there are  $(9)_2 = \underbrace{9 \cdot 8}_{\text{there are 2 terms}} = 72$  possible dispositions.

3. How many 6-digit phone numbers can we form if each digit is used once?

**Answer:** there are  $(10)_6 = \underbrace{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}_{\text{il y a 6 termes}} = 151200$  possible phone numbers.

## 1.4 Permutation (without repetition)

**Definition 1.4.1.** We call a *permutation* without repetition (we often do not specify *without repetition*) of  $n$  elements of  $E$ , any sequence of length  $n$  formed by the elements of  $E$ , (or it is an ordered disposition of the  $n$  elements of  $E$ ), the number of permutations is denoted by  $\mathcal{P}_n$ .

In a such permutation (without repetition), each one elements of  $E$  appears once, in a single fixed order.

**Theorem 1.4.2.** We have

$$\mathcal{P}_n = n! \quad (1.4.1)$$

*Proof.*

**1<sup>st</sup> approach:**

A permutation (without repetition) is an arrangement of  $n$  elements, then

$$\mathcal{P}_n = (n)_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n!.$$

**2<sup>nd</sup> approach:** (by induction on  $n$ )

Let  $x \in E$ , in  $E \setminus \{x\}$  there are  $(n-1)$  elements so  $\mathcal{P}_{n-1}$  permutations are formed using the  $(n-1)$  elements of  $E \setminus \{x\}$ . From each of theses  $(n-1)$  permutations, we can form  $n$  permutations, just place  $x$  at one of the  $n$  possible positions, namely before the 1<sup>st</sup> element, between the 1<sup>st</sup> and the 2<sup>nd</sup> element,..., after the  $(n-1)$ <sup>st</sup> element. So  $\mathcal{P}_n = n\mathcal{P}_{n-1}$ , in the same way  $\mathcal{P}_{n-1} = (n-1)\mathcal{P}_{n-2}, \dots, \mathcal{P}_2 = 2\mathcal{P}_1$  where  $\mathcal{P}_n = n \cdot (n-1) \cdots 2 \cdot \mathcal{P}_1$  since  $\mathcal{P}_1 = 1$  (number of dispositions with unique element), it results  $\mathcal{P}_n = n!$ .  $\square$

**Property 1.4.3.** We have, for all integer  $k \in \{1, 2, \dots, n\}$

$$\mathcal{P}_n = (n)_k \mathcal{P}_{n-k} \quad (1.4.2)$$

*Proof.* **1<sup>st</sup> approach** (direct calculation):

$$(n)_k \mathcal{P}_{n-k} = \frac{n!}{(n-k)!} (n-k)! = n! = \mathcal{P}_n.$$

**2<sup>nd</sup> approach** (combinatorial argument): A permutation of  $n$  elements is an arrangement of  $k$  elements taken from  $n$ , and for each one of the  $k$  elements we have a permutation of  $(n-k)$  remaining elements, so by fundamental principal of counting, we get

$$\mathcal{P}_n = \mathcal{P}_{n-k} \cdot (n)_k.$$

$\square$

#### Example 1.4.4.

1. In how many ways could we arrange 6 persons around a table if this table: i. has the shape  $\cup$ ?  
ii. is round?

**Answer:**

- i. There are as many ways of permuting 6 elements:  $\mathcal{P}_6 = 6! = 720$ .
  - ii. If the table is round, we place the 1<sup>st</sup> person at any location, this leads us to the previous problem with the 5 remaining persons, so the number of ways of arranging them is  $\mathcal{P}_5 = 5! = 120$ .
2. The course of DM is taken by 5 technologists and 3 computer scientists. In the exam, they are classified according to their marks. (it is assumed that no two students have the same mark).
    - i. How many rankings can we have?
    - ii. If the technologists are classified among themselves and the computer scientists are classified among themselves, how many global rankings can we have?

**Answer:**

- i. It is the number of permutations of 8 students, so the number of rankings of students is  $\mathcal{P}_8 = 8! = 40320$ .
  - ii. Since there are  $\mathcal{P}_5 = 5!$  rankings of technologists between themselves and  $\mathcal{P}_3 = 3!$  rankings of computer scientists between themselves. So by FP, there are  $\mathcal{P}_5 \cdot \mathcal{P}_3 = 5!3! = 720$  possible rankings.
3. The librarian will arrange 15 books on a shelf of the library, 4 of them are maths books, 6 statistics books, 3 physics books and 1 chemistry book. The librarian would like to arrange the books so that all those dealing with the same subject remain in a group. How many possible dispositions are there?

**Answer:**

There are:  $\mathcal{P}_4$  permutations of maths books among themselves,  $\mathcal{P}_6$  permutations of statistics books among themselves,

$\mathcal{P}_3$  permutations of physics books among themselves,

$\mathcal{P}_1$  permutations of chemistry book,

by the generalized fundamental principle of counting, there are  $\mathcal{P}_4 \cdot \mathcal{P}_6 \cdot \mathcal{P}_3 \cdot \mathcal{P}_1$  dispositions for arranging the maths books after that statistics books, then physics books, and then chemistry book. There are  $\mathcal{P}_4$  ways to permute the 4 groups of books among themselves. So the desired answer is:  $\mathcal{P}_4 (\mathcal{P}_4 \cdot \mathcal{P}_6 \cdot \mathcal{P}_3 \cdot \mathcal{P}_1) = 4! \cdot 4! \cdot 6! \cdot 3! \cdot 1! = 2488320$  possible dispositions.

## 1.5 Combination (without repetition)

**Definition 1.5.1.** We call **combination** without repetition (we often do not specify **without repetition**) of  $n$  elements, all subsets of  $E$  of cardinal  $k$ , (or it is an unordered disposition without repetition,

of  $k$  elements chosen from the  $n$  of  $E$ ), the number of combinations is denoted by  $\binom{n}{k}$  (or  $C_n^k$ ).  $\binom{n}{k}$  are called **binomial coefficients**.

In a such combination (without repetition), each one of  $k$  elements appears once, so  $k \in \{1, 2, \dots, n\}$ .

**Theorem 1.5.2.** *We have*

$$\binom{n}{k} = \frac{(n)_k}{k!}. \quad (1.5.1)$$

*Proof.* The number of ways to choose  $k$  elements from  $n$  is  $\binom{n}{k}$ , if we also arrange each group of the  $k$  obtained elements, we have  $k!$  ways of doing so (because it is a permutation). So by FP, the number of arrangements of  $k$  elements chosen from  $n$  is  $\mathcal{P}_k \cdot \binom{n}{k}$ , which means:  $(n)_k = \mathcal{P}_k \binom{n}{k}$ , hence the result.  $\square$

**Remark 1.5.3.** *As consequence of Remark 3.3, we can extend the definition of  $\binom{n}{k}$  for  $k \geq 0$ :*

$$\binom{n}{k} = \begin{cases} 1, & \text{if } k = 0, \\ \frac{(n)_k}{k!}, & \text{if } k = 1, 2, \dots, n, \\ 0, & \text{if } k > n. \end{cases} \quad (1.5.2)$$

so

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} \quad k = 0, 1, 2, \dots, n. \quad (1.5.3)$$

**Consequence 1.5.4.**

$$\binom{n}{k} = \binom{n}{n-k} \text{ and so } \binom{n}{0} = \binom{n}{n} = 1 \text{ and } \binom{n}{1} = \binom{n}{n-1} = n.$$

**Property 1.5.5.**

$$\binom{n}{k} = \frac{n}{k} \binom{n-1}{k-1}, \quad (1.5.4)$$

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1} \quad \text{Pascal's formula.} \quad (1.5.5)$$

*Proof.*

$$1. \binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n}{k} \frac{(n-1)!}{[(n-1)-(k-1)]!(k-1)!} = \frac{n}{k} \binom{n-1}{k-1}.$$

$$\begin{aligned} 2. \binom{n-1}{k} + \binom{n-1}{k-1} &= \frac{(n-1)!}{k!(n-k-1)!} + \frac{(n-1)!}{(n-k)!(k-1)!} = \frac{(n-1)!(n-k)}{(n-k)!k!} + \frac{(n-1)!k}{(n-k)!k!} \\ &= \frac{(n-1)!(n-k+k)}{(n-k)!k!} = \binom{n}{k}. \end{aligned}$$

$\square$

**Remark 1.5.6.** We prove the theorem 1.5.2 using a combinatorial argument, we could also prove it using (1.5.5) and reasoning by induction. But in this case, the proof of (1.5.5) must be done using a combinatorial argument (which will be the subject of an exercise).

**Example 1.5.7.**

1. We want to form a committee of 3 students from the 15 postgraduate students, How many committees can we form? **Answer:** There are  $\binom{15}{3} = \frac{15 \cdot 14 \cdot 13}{1 \cdot 2 \cdot 3} = 455$  possible committees.
2. An urn contains: 5 white balls (*wb*) numbered from 1 to 5 ,  
3 black balls (*bb*) numbered from 1 to 3,  
2 green balls(*gb*) numbered from 1 to 2,  
and 1 yellow ball (*yb*) with the number 1 .  
How many 5-ball draws are there if
  - i. there is no restriction?
  - ii. there are exactly 2 white balls?
  - iii. at least two balls are black?
  - iv. at most one ball is white?
  - v. there are exactly 2*wb* 2*bb* 1*gb*?
  - vi. the green and the black balls can't be together?
  - vii. two balls having the  $b \neq 2$ , one having the  $n \neq 3$  and one having the  $n \neq 5$ ?
  - viii. two *wb*, 1*yb* and two balls having the  $n \neq 2$ ?

**Answer:**

- i. There are  $\binom{11}{5} = \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 462$  possible draws.
- ii. There are  $\binom{5}{2}$  possibilities to choose two *wb* from the 5 *wb* and  $\binom{6}{3}$  possibilities to choose three from the rest and this for each choice of the 1<sup>st</sup> case, so by (FP), there are  $\binom{5}{2} \cdot \binom{6}{3} = \frac{5 \cdot 4 \cdot 6 \cdot 5 \cdot 4}{2 \cdot 1 \cdot 3 \cdot 2 \cdot 1} = 280$  possible draws.
- iii. Having at least two *bb*, it is either having exactly 2 *bb* or having exactly 3 *bb*. So it is the number of draws where we have exactly two *bb* plus the number of draws where we have exactly three *bb* (see ii.)  $\binom{3}{2} \binom{8}{3} + \binom{3}{3} \binom{8}{2} = \frac{3 \cdot 2 \cdot 8 \cdot 7 \cdot 6}{2 \cdot 1 \cdot 3 \cdot 2 \cdot 1} + 1 \cdot \frac{8 \cdot 7}{2 \cdot 1} = 196$  draws.
- iv. Having at most 1 *wb* means either having exactly 0 *wb* or having exactly 1 *wb*. Which gives (see iii.)  $\binom{5}{0} \binom{6}{5} + \binom{5}{1} \binom{6}{4} = 1 \cdot 6 + 5 \cdot \frac{6 \cdot 5 \cdot 4 \cdot 3}{4 \cdot 3 \cdot 2 \cdot 1} = 81$  draws. Or, it is the

total 5- balls draw minus the number of draws having at least 2  $wb$ , which gives  $\binom{11}{5} - \left[ \binom{5}{2} \binom{6}{3} + \binom{5}{3} \binom{6}{2} + \binom{5}{4} \binom{6}{1} + \binom{5}{5} \binom{6}{0} \right]$ .

v. There are  $\binom{5}{2}_{wb} \binom{3}{2}_{bb} \binom{2}{1}_{gb} \binom{1}{0}_{yb} = 10 \cdot 3 \cdot 2 \cdot 1 = 60$  possible draws.

vi. This is equivalent to: (not having  $gb$ ) or (having  $gb$  and not having  $bb$ ) this gives  $\binom{2}{0}_{bv} \binom{9}{5}_R +$

$$\binom{3}{0}_{bn} \left[ \binom{2}{2}_{bv} \binom{6}{5}_R + \binom{2}{1}_{bv} \binom{6}{4}_R \right] = 1 \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} + 1 \cdot \left[ 1 \cdot 6 + 2 \cdot \frac{6 \cdot 5}{2 \cdot 1} \right] = 162.$$

vii. There are 4 balls with  $n^{\circ}1$ , 3 balls with  $n^{\circ}2$ , 2 balls with  $n^{\circ}3$ , 1 ball with  $n^{\circ}4$  and 1 ball with  $n^{\circ}5$ . So we have another subdivision of the 11 balls (we go back to the reasoning of question v.) which gives  $\binom{4}{0}_{n^{\circ}1} \binom{3}{2}_{n^{\circ}2} \binom{2}{1}_{n^{\circ}3} \binom{1}{0}_{n^{\circ}4} \binom{1}{1}_{n^{\circ}5} = 1 \cdot 3 \cdot 2 \cdot 1 \cdot 1 = 6$  possible draws.

viii. The subdivision of the 11 balls is redone as follows:

1. there are 4  $wb$  which does not have the  $n^{\circ}2$ ,
2. there is 1  $wb$  having the  $n^{\circ}2$ ,
3. there are 2 balls (which are not white) and have the  $n^{\circ}2$ ,
4. there is 1  $yb$ ,
5. there are 3 remaining balls.

There are two possibilities: choosing the  $wb$   $n^{\circ}2$  or not, this gives:

$$\binom{4}{1}_1 \binom{1}{1}_2 \binom{2}{1}_3 \binom{1}{1}_4 \binom{3}{1}_5 + \binom{4}{2}_1 \binom{1}{0}_2 \binom{2}{2}_3 \binom{1}{1}_4 \binom{3}{0}_5.$$

### 1.5.1 Newton binomial formula

**Theorem 1.5.8** (Binomial theorem). *For all complex numbers  $a$  and  $b$  and for all integers  $n \geq 0$ , we have*

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}.$$

*Proof.*

**1<sup>st</sup> solution:** by induction on  $n$  (we use the Pascal's formula).

**2<sup>nd</sup> solution:** using a combinatorial argument. Let consider the product  $(a_1 + b_1)(a_2 + b_2) \cdot \dots \cdot (a_n + b_n)$ , by developing it, we get the sum of  $2^n$  terms, each term being the product of  $n$  factors. Each one of  $2^n$  terms contains for all  $i = 1, 2, \dots, n$ ; even the factor  $a_i$  or factor  $b_i$ .

(example  $(a_1 + b_1)(a_2 + b_2)(a_3 + b_3) = a_1a_2a_3 + a_1a_2b_3 + a_1b_2a_3 + b_1a_2a_3 + b_1b_2b_3 + b_1b_2a_3 + b_1a_2b_3 + a_1b_2b_3$ ), If now, we put  $a_i = a$  and  $b_i = b \quad i = 1, \dots, n$  there are  $\binom{n}{k}$  which contains  $a^k$  (and  $\binom{n}{n-k}$  terms containing  $b^{n-k}$ )  
so

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^k b^{n-k}$$

□

Pascal triangle (consequence Pascal's formula)

$$\begin{array}{ccccccccc} & & \binom{0}{0} & & & & & & 1 \\ & & \binom{1}{0} & \binom{1}{1} & & & & & 1 & 1 \\ & & \binom{2}{0} & \binom{2}{1} & \binom{2}{2} & & & & 1 & 2 & 1 \\ & & \binom{3}{0} & \binom{3}{1} & \binom{3}{2} & \binom{3}{3} & & & 1 & 3 & 3 & 1 \\ & & \binom{4}{0} & \binom{4}{1} & \binom{4}{2} & \binom{4}{3} & \binom{4}{4} & & 1 & 4 & 6 & 4 & 1 \\ & & \binom{5}{0} & \binom{5}{1} & \binom{5}{2} & \binom{5}{3} & \binom{5}{4} & \binom{5}{5} & 1 & 5 & 10 & 10 & 5 & 1 \end{array}$$

$$(a + b)^0 = a^0 b^0$$

$$(a + b)^1 = 1a^1 b^0 + 1a^0 b^1$$

$$(a + b)^2 = 1a^2 b^0 + 2a^1 b^1 + 1a^0 b^2$$

$$(a + b)^3 = 1a^3 b^0 + 3a^2 b^1 + 3a^1 b^2 + 1a^0 b^3$$

$$(a + b)^4 = 1a^4 b^0 + 4a^3 b^1 + 6a^2 b^2 + 4a^1 b^3 + 1a^0 b^4$$

$$(a + b)^5 = 1a^5 b^0 + 5a^4 b^1 + 10a^3 b^2 + 10a^2 b^3 + 5a^1 b^4 + 1a^0 b^5$$

**Example 1.5.9.** How many subsets are there for a set of  $n$  elements?

**Answer:** There are  $\binom{n}{0}$  empty subsets,  $\binom{n}{1}$  subsets having 1 element, ...,  $\binom{n}{n}$  subsets having  $n$  elements, so, in total there are  $\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n} = \sum_{k=0}^n \binom{n}{k} = (1 + 1)^n = 2^n$  subsets.

## 1.6 Permutation with repetition

**Definition 1.6.1.** We call **permutation with repetition** of  $n$  elements formed from  $r$  discernible groups of indiscernible elements, an ordered disposition of the set of  $n$  elements, in the 1<sup>st</sup> group there are  $n_1$  indiscernible elements, in the 2<sup>nd</sup> group there are  $n_2$ , ..., and in the  $r$ <sup>st</sup> group there are  $n_r$ , and the  $r$  groups are discernible (or it is the number of possible distribution of  $n$  objects on  $r$  distinct groups

of size  $n_1, n_2, \dots, n_r$  respectively),  $n = n_1 + n_2 + \dots + n_r = \sum_{i=1}^r n_i$ , we denote this number by  $\binom{n}{n_1, n_2, \dots, n_r}$ , it is called **multinomial coefficient**.

In a such permutation (with repetition), the 1<sup>st</sup> element appears  $n_1$  times, the 2<sup>nd</sup> element appears  $n_2$  times,..., the  $r$ <sup>st</sup> element appears  $n_r$  times, in a single fixed order.

We can draw a diagram of  $E$ , as follows:  $\overbrace{(a, \dots, a) (b, \dots, b) \dots (t, \dots, t)}^{n \text{ terms}}$   
 $\underbrace{\hspace{1.5cm}}_{n_1 \text{ times}} \underbrace{\hspace{1.5cm}}_{n_2 \text{ times}} \underbrace{\hspace{1.5cm}}_{n_r \text{ times}}$   
 $r \leq n \quad n_1 + n_2 + \dots + n_r = n$  (with all the possible "disorders")

**Theorem 1.6.2.**

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1! n_2! \dots n_r!}. \quad (1.6.1)$$

*Proof.* We consider all possible permutations of the  $n$  elements: there are  $\mathcal{P}_n$ , but the  $n_1$  elements of the 1<sup>st</sup> group are already counted  $\mathcal{P}_{n_1}$  times (because they were initially considered to be discernible, whereas they are indiscernible, then the  $\mathcal{P}_{n_1}$  dispositions are really only one), same thing for the  $n_2$  elements of 2<sup>nd</sup> group have been counted  $\mathcal{P}_{n_2}$  times,..., the  $n_r$  elements of  $r$ <sup>th</sup> group have been counted  $\mathcal{P}_{n_r}$  times. Then, to consider all possible permutations we have to divide  $\mathcal{P}_n$  by  $\mathcal{P}_{n_1}$ , by  $\mathcal{P}_{n_2}$ ,..., by  $\mathcal{P}_{n_r}$  so the number of permutations with repetition is:

$$\binom{n}{n_1, \dots, n_r} = \frac{\mathcal{P}_n}{\mathcal{P}_{n_1} \mathcal{P}_{n_2} \dots \mathcal{P}_{n_r}} = \frac{n!}{n_1! n_2! \dots n_r!}.$$

□

**Exercise 1.6.3.** Redo the proof using the generalized fundamental principal.

**Consequence 1.6.4.**

1. The permutation of the  $r$  groups among themselves, doesn't change the final result, example  $\binom{n}{p, q, r} = \binom{n}{p, r, q} = \binom{n}{q, r, p} = \binom{n}{q, p, r} = \binom{n}{r, p, q} = \binom{n}{r, q, p} = \frac{n!}{p! q! r!}$ .
2.  $\binom{n}{n_1, n_2, \dots, n_r} \leq \mathcal{P}_n$  and it is equal if and only if  $n = r \Leftrightarrow n_1 = \dots = n_r = 1$ .

**Property 1.6.5.**

1. Relation between multinomial coefficient and binomial coefficients:

$$\binom{n}{n_1, n_2, \dots, n_r} = \binom{n}{n_1} \binom{n - n_1}{n_2} \dots \binom{n - n_1 - \dots - n_{r-1}}{n_r} = \prod_{i=1}^r \binom{n - \sum_{j=1}^{i-1} n_j}{n_i} \quad (1.6.2)$$

- 2.

$$\begin{aligned} \binom{n}{n_1, n_2, \dots, n_r} &= \binom{n-1}{n_1-1, n_2, \dots, n_r} + \binom{n-1}{n_1, n_2-1, \dots, n_r} + \dots + \binom{n-1}{n_1, n_2, \dots, n_r-1} \\ &= \sum_{i=1}^r \binom{n-1}{n_1, \dots, n_{i-1}, n_i-1, \dots, n_r} \end{aligned} \quad (1.6.3)$$

*Proof.*

$$\begin{aligned}
1. & \binom{n}{n_1} \binom{n-n_1}{n_2} \cdots \binom{n-n_1-\cdots-n_{r-1}}{n_r} \\
&= \frac{n!}{n_1!(n-n_1)!} \frac{(n-n_1)!}{n_2!(n-n_1-n_2)!} \cdots \frac{(n-n_1-\cdots-n_{r-1})!}{n_r! \underbrace{(n-n_1-\cdots-n_{r-1}-n_r)!}_{=0}} \\
&= \frac{n!}{n_1!n_2! \cdots n_r!} = \binom{n}{n_1, \dots, n_r}. \\
2. & \binom{n-1}{n_1-1, n_2, \dots, n_r} + \binom{n-1}{n_1, n_2-1, \dots, n_r} + \cdots + \binom{n-1}{n_1, n_2, \dots, n_r-1} \\
&= \frac{(n-1)!}{(n_1-1)!n_2! \cdots n_r!} + \frac{(n-1)!}{n_1!(n_2-1)! \cdots n_r!} + \cdots + \frac{(n-1)!}{n_1!n_2! \cdots (n_r-1)!} \\
&= \frac{(n-1)!n_1}{n_1!n_2! \cdots n_r!} + \frac{(n-1)!n_2}{n_1!n_2! \cdots n_r!} + \cdots + \frac{(n-1)!n_r}{n_1!n_2! \cdots n_r!} \\
&= \frac{(n-1)!}{n_1!n_2! \cdots n_r!} \underbrace{(n_1 + n_2 + \cdots + n_r)}_n = \frac{n!}{n_1!n_2! \cdots n_r!} = \binom{n}{n_1, n_2, \dots, n_r}.
\end{aligned}$$

□

**Remark 1.6.6** (Combination and permutation with repetition). *A combination is a permutation with repetition of  $n$  elements formed by two groups of  $k$  and  $(n - k)$  elements respectively, then*

$$\binom{n}{k} = \binom{n}{n-k} = \frac{\mathcal{P}_n}{\mathcal{P}_k \mathcal{P}_{n-k}}. \quad (1.6.4)$$

**Example 1.6.7.**

1. From the following 6 numbers 5,5,5,2,2,9. How many phone number can we form?

**Answer:** it is a permutation with repetition, with  $n_1 = 3$ ,  $n_2 = 2$ ,  $n_3 = 1$ , so the number of combination is  $\binom{6}{3,2,1} = \binom{6}{2,1,3} = \binom{6}{2,3,1} = \binom{6}{3,1,2} = \binom{6}{1,2,3} = \binom{6}{1,3,2} = \frac{6!}{3!2!1!} = 60$

|        |         |        |        |        |        |        |        |        |        |
|--------|---------|--------|--------|--------|--------|--------|--------|--------|--------|
| 555229 | 552529  | 552259 | 552295 | 525295 | 525259 | 525529 | 522559 | 512955 | 229555 |
| 555292 | 552592  | 552952 | 552925 | 525925 | 525952 | 525592 | 529552 | 529255 | 292555 |
| 555922 | 559522  | 559252 | 559225 | 595225 | 595252 | 595522 | 592552 | 592255 | 922555 |
| 522595 | 255529  | 255259 | 255295 | 225595 | 252595 | 252559 | 252955 | 225559 | 225955 |
| 529525 | 255592  | 255152 | 255925 | 295525 | 259525 | 259552 | 259255 | 295552 | 295255 |
| 592525 | 95552.2 | 955252 | 955225 | 925525 | 952525 | 952552 | 952255 | 925552 | 925255 |

2. Among the 15 participants in a competition for a scholarship there are 6 Algerians, 3 Moroccans, 3 Tunisian, 2 Libyans et 1 Mauritanian. If the competition ranking lists the nationalities of

competitors (not their identities), how many different rankings does such list correspond to?

**Answer:** it is a permutation with repetition; with  $n_1 = 6$ ,  $n_2 = n_3 = 3$ ,  $n_4 = 2$  et  $n_5 = 1$  ( $r = 5$ ) ( $n = n_1 + n_2 + n_3 + n_4 + n_5 = 15$ ), so the number of individual rankings is  $\binom{15}{6, 3, 3, 2, 1} = \frac{15!}{6!3!3!2!1!} = 25225200$

### 1.6.1 Multinomial Newton formula

**Theorem 1.6.8** (Multinomial coefficient). For all  $a_1, a_2, \dots, a_r \in \mathbb{C}$  and all  $n \in \mathbb{N} \cup \{0\}$  we have

$$(a_1 + a_2 + \dots + a_r)^n = \sum_{n_1, n_2, \dots, n_r / \sum_{k=1}^r n_k = n} \binom{n}{n_1, n_2, \dots, n_r} a_1^{n_1} a_2^{n_2} \dots a_r^{n_r}.$$

*Proof.* It can be shown in three different ways:

1<sup>st</sup> **approach:** by induction on  $n$  using the relation (1.6.3).

2<sup>nd</sup> **approach:** using a combinatorial argument (by generalizing Newton's binomial)

3<sup>rd</sup> **approach:** using Newton's binomial and the relation (1.6.2) which will be presented. (it is an induction on  $r$ )

$$\begin{aligned} (a_1 + a_2 + \dots + a_r)^n &= [a_1 + (a_2 + \dots + a_r)]^n \\ &= \sum_{n_1=0}^n \binom{n}{n_1} a_1^{n_1} (a_2 + \dots + a_r)^{n-n_1} \\ &= \sum_{n_1=0}^n \binom{n}{n_1} a_1^{n_1} \left\{ \sum_{n_2=0}^{n-n_1} \binom{n-n_1}{n_2} a_2^{n_2} (a_3 + \dots + a_r)^{n-n_1-n_2} \right\} \\ &\vdots \\ &= \sum_{n_1=0}^n \sum_{n_2=0}^{n-n_1} \dots \sum_{n_r=0}^{n-n_1-\dots-n_{r-1}} \binom{n}{n_1} \binom{n-n_1}{n_2} \dots \binom{n-n_1-\dots-n_{r-1}}{n_r} a_1^{n_1} a_2^{n_2} \dots a_r^{n_r} \end{aligned}$$

from (1.6.2)

$$= \sum_{n_1, n_2, \dots, n_r / \sum_{j=1}^r n_j = n} \binom{n}{n_1, n_2, \dots, n_r} a_1^{n_1} a_2^{n_2} \dots a_r^{n_r}$$

□

**Example 1.6.9.**

$$\begin{aligned} (a + b + c)^2 &= \sum_{i, j, k / i+j+k=2} \binom{2}{i, j, k} a^i b^j c^k \\ &= \binom{2}{2, 0, 0} a^2 b^0 c^0 + \binom{2}{0, 2, 0} a^0 b^2 c^0 + \binom{2}{0, 0, 2} a^0 b^0 c^2 \\ &\quad + \binom{2}{1, 1, 0} a^1 b^1 c^0 + \binom{2}{1, 0, 1} a^1 b^0 c^1 + \binom{2}{0, 1, 1} a^0 b^1 c^1 \\ &= a^2 + b^2 + c^2 + 2ab + 2ac + 2bc. \end{aligned}$$

## 1.7 Combination with repetition

Soit  $E = \{a_1, a_2, \dots, a_n\}$ .

**Definition 1.7.1.** We call **combination with repetition** of  $n$  elements, the sequence where the  $k_1$  first terms are equals to  $a_1$ , the  $k_2$  next ones are equals to  $a_2, \dots$ , the  $k_n$  last ones are equals to  $a_n$  with  $k_1 + k_2 + \dots + k_n = k$  (or it is a unordered disposition, with repetition of  $k$  elements chosen from the  $n$  elements of  $E$ ). We denote this number by  $\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right)$ .

In such combination with repetition some of the  $k$  elements can appear many times (even  $k$  times), so  $k \in \mathbb{N} \cup \{0\}$ .

**Theorem 1.7.2.**

$$\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right) = \binom{n+k-1}{n-1}. \quad (1.7.1)$$

*Proof.* As the order is not involved in a combination with repetition, from the  $k$  chosen elements, we can consider that we have choose:  $k_1$  times  $a_1$  then a separator, then  $k_2$  times  $a_2$  then a separator, ..., then  $k_n$  times  $a_n$ .

$$\underbrace{a_1 \cdots a_1}_{k_1 \text{ times}} \mid \underbrace{a_2 \cdots a_2}_{k_2 \text{ times}} \mid \cdots \mid \underbrace{a_n \cdots a_n}_{k_n \text{ times}}.$$

separator                      separator

In all, there are  $(n-1)$  separators and  $k_1 + k_2 + \dots + k_n = k$  elements, and the  $(n-1)$  separators have  $\binom{n+k-1}{n-1}$  possibilities to be placed in the  $(n+k-1)$  positions (it is a combination (without repetition) of  $(n-1)$  elements taken from  $(n+k-1)$ ) where  $\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right) = \binom{n+k-1}{n-1}$ . □

**Remark 1.7.3.** We can extend the definition of  $\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right)$  for  $k = 0$  by  $\left(\left(\begin{smallmatrix} n \\ 0 \end{smallmatrix}\right)\right) = 1 (= \binom{n}{0})$  and then

$$\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right) = \frac{(n+k-1)!}{(n-1)!k!}. \quad (1.7.2)$$

**Consequence 1.7.4.**

1.  $\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right) = \binom{n+k-1}{k},$
2.  $\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right) = \left(\left(\begin{smallmatrix} n-1 \\ k+1 \end{smallmatrix}\right)\right),$
3.  $\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right) \geq \binom{n}{k}$  and  $\left(\left(\begin{smallmatrix} n \\ k \end{smallmatrix}\right)\right) = \binom{n}{k} \Leftrightarrow k = 1.$

**Example 1.7.5.**

1. In an urn there are 6 indiscernible balls, we draw 3, one by one by handing them over each time. How many possibilities are there?

**Answer:** it is a combination with repetition of 3 elements taken from 6, so there are  $\left(\binom{6}{3}\right) = \binom{6+3-1}{3} = \binom{8}{3} = 56$  possibilities.

2. 8 students making (each) 3 draws, to decide who will borrow the books available in the library. How many possible combinations are there?

**Answer:** Whoever borrows one book, can borrow the other two (or another one). So, it is the number of combinations with repetition of 8 elements 3 to 3. ie:  $\left(\binom{8}{3}\right) = \binom{3+8-1}{3} = \binom{10}{3} = 120$  possible combinations. Each combination is the borrowing of 3 different students or not. [It should be noted that if we were in the case where a student cannot take more than one book, we should be in the situation of a combination (without repetition) of 8 elements taken 3 to 3. ie:  $\binom{8}{3} = 56 << 120$ ]

3. How many terms are there in the development of  $(a_1 + a_2 + \dots + a_n)^m$ ?

**Answer:** it is the number of combinations with repetition of  $m$  elements chosen from  $r$ , indeed

$$(a_1 + a_2 + \dots + a_r)^m = \sum_{n_1, n_2, \dots, n_r / \sum_{j=1}^r n_j = m} \binom{m}{n_1, \dots, n_r} a_1^{n_1} a_2^{n_2} \dots a_r^{n_r}.$$

In  $a_1^{n_1} \dots a_r^{n_r}$  there are  $n$  elements of  $\{a_1, \dots, a_n\}$  and we take  $n_1$  times the element  $a_1$ ,  $n_2$  times the element  $a_2$ , ...,  $n_r$  times the element  $a_r$ , with all possible combinations, so there are  $\left(\binom{r}{m}\right)$ . [In the example (1.6.9) there  $\left(\binom{3}{2}\right) = 6$  terms (which the case)].

This last example allows us to show the following result:

**Theorem 1.7.6.** The equation  $x_1 + x_2 + \dots + x_k = n$  has exactly  $\left(\binom{k}{n}\right)$  distinct solutions in  $\mathbb{N} \cup \{0\}$ .

*Proof.* This is exactly the summation index of the multinomial formula. □

**Corollary 1.7.7.** The equation  $x_1 + x_2 + \dots + x_k = n$  has exactly  $\binom{n-1}{k-1}$  distinct solutions in  $\mathbb{N}$ .

*Proof.* Just do the change of variables  $y_i = x_i - 1$  and we get  $y_1 + y_2 + \dots + y_k = n - k$  with  $y_i \in \mathbb{N} \cup \{0\}$ , then, by Theorem 1.7.6 there are  $\left(\binom{k}{n-k}\right) = \binom{n-1}{k-1}$  solutions. □

**Example 1.7.8.**

How many solutions does the equation  $x + y = 4$  have in i.  $\mathbb{N} \cup \{0\}$  ii.  $\mathbb{N}$ .

**Answer:**

i. In  $\mathbb{N} \cup \{0\}$  there are  $\left(\binom{2}{4}\right) = \frac{5!}{4!1!}$  solutions, which are  $(0, 4); (1, 3); (2, 2); (3, 1); (4, 0)$ .

ii. In  $\mathbb{N}$  there are  $\binom{3}{1} = 3$  solutions, which are:  $(1, 3); (2, 2); (3, 1)$ .